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Original Article

Tools for investigating host-microbe crosstalk using glycan analysis probes inspired by human lectins

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Human lectins are critical carbohydrate-binding proteins that recognize diverse glycoconjugates from microorganisms and can play a key role in host–microbe interactions. Despite their importance in immune recognition and microbe binding, the specific glycan ligands and functions of many human lectins remain poorly understood. Using previous proof-of-concept studies on selected lectins as the foundation for this work, we present ten additional glycan analysis probes (GAPs) from a diverse set of human soluble lectins, offering robust tools to investigate glycan-mediated interactions. We describe a protein engineering platform that enables scalable production of GAPs that maintain native-like conformations and oligomerization states, equipped with functional reporter tags for targeted glycan profiling. We demonstrate that the soluble GAP reagents can be used in various applications, including glycan array analysis, mucin-binding assays, tissue staining, and microbe binding in complex populations. These capabilities make GAPs valuable for dissecting interactions relevant to understanding host responses to microbes. The tools can also be used to probe differential microbial and mammalian glycan interactions, which are crucial for understanding the interactions of lectins in a physiological environment where both glycan types exist. GAPs have potential as diagnostic and prognostic tools for detecting glycan alterations in chronic diseases, microbial dysbiosis, and immune-related conditions.

Key words: glycan array; host–microbe interactions; microbiome analysis; microbial glycans.

Introduction

Human soluble lectins (Sahly et al. 2008; Raposo et al. 2021; Schnider et al. 2023) are carbohydrate-binding proteins in the extracellular matrix that play critical roles in host–microbe interactions. These lectins have evolved to recognize distinct glycan structures and drive various physiological functions, including pathogen recognition and innate immune functions. The biological significance of these human lectins has prompted studies on their specificity towards N- and O-linked mammalian glycans (Kletter et al. 2013). Still, the glycan and microbe-binding specificities of many remain unclear. Addressing this incomplete understanding of the binding profiles of soluble human lectins towards microbial glycan targets has been difficult using current tools such as glycan arrays, as microbial cell surface glycans comprise remarkably diverse monosaccharide building blocks and glycosidic linkages absent in the mammalian glycome (Herget et al. 2008; Adibekian et al. 2011; Imperiali 2019; Griffin and Hsieh-Wilson 2022). As human lectins serve as "readers" of complex microbial surface glycans, they present opportunities for developing reagents for understanding host cell recognition, as well as bacterial colonization and pathogenesis.

Recently, we established a proof-of-concept for producing and utilizing glycan analysis probes (GAPs) from soluble human lectins to decode host specificity to microbes within complex microbiomes (Ghosh et al. 2023; McPherson et al. 2023). GAPs represent soluble protein reagents that can be engineered with fluorophores or affinity handles; thus, they can be applied to specifically bind to microbial cell surface glycans, enabling their detection in mixed microbial populations and enrichment for analysis and characterization. GAPs provide capabilities that are complementary to lectin arrays (Hsu and Mahal 2006; Benjamin et al. 2024). For example, we demonstrated that the GAP based on the human lectin Zymogen granule protein ZG16B selectively binds to a subset of commensal bacteria in the oral cavity, regulating their growth by recruiting oral mucin MUC7 to maintain homeostasis (Ghosh et al. 2023). The ZG16B-GAP also allowed the identification of the glycan epitope on cell wall polysaccharide on the targeted bacteria involved in the lectin-glycan interaction. In another example, mannose-binding lectin (MBL) and intelectin-1 (hItln1) GAPs were shown to bind distinct microbial populations in stool samples from healthy individuals and those with inflammatory bowel disease (McPherson et al. 2023), suggesting potential applications of GAPs as diagnostic markers for diseases associated with microbial dysbiosis.

Building on these studies we have expanded the suite of GAPs to include an additional ten human soluble lectins, presenting new resources for investigating and understanding human lectin function and providing insights into microbial glycan diversity and glycan-mediated host–microbe interactions. Here, we detail the modular platform (Fig. 1A) for

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tailoring soluble human lectins as glycan analysis probes (GAPs) for visualizing the glycan-binding properties to gain insight into their biological functions. We selected a series of human lectins based on tissue distribution (Fig. 1B), serum abundance (Fig. 1B), protein structure (Fig. 1C), and functional diversity as candidates for conversion into functionalized GAPs. The generalized approach provides lectins bearing a range of functional tags in high yield and purity for use in biophysical and biochemical assays. An essential feature of the GAP production pipeline is its compatibility with newly targeted lectins, enabling rapid screening of expression and purification conditions and co-expression tag optimization.

Our comparative analysis reveals the distinct binding profiles of lectin-based GAPs as assessed by microbial and mammalian glycan arrays and shows applications of the GAPs in mucin binding and tissue staining analyses. As seen with the ZG16B-GAP, such applications are particularly useful for exploring ancillary lectin interactions with host glycoproteins, which play a crucial role in mucosal immunity (Holmgren and Czerkinsky 2005). Additionally, different GAPs exhibited distinct binding patterns to microbial populations in fecal samples from healthy and diseased individuals, demonstrating their ability to recognize gut-resident species that can contribute to human health within a native community. We present GAPs as valuable resources that can be employed to delineate the role of lectins in physiologically-relevant environments in host–microbe interactions.

Results

Selection of representative lectins as targets for GAP development

We describe the development of a curated set of GAPs from soluble human lectins based on reported microbe binding and participation in innate immunity, as shown in Table 1. The lectins feature different tissue-specific distributions (Fig. 1B), abundance, oligomeric states, potential ligand-specificity, and reported biological functions. With regard to function, we focused on soluble lectins involved in innate immunity with different roles (Table 1). Some can bind and kill microbes (e.g. the galectins); others can restrict microbial growth and promote microbe removal (e.g. ZG16P, SP-D), while yet others are involved in complement-mediated immunity (MBL, CRP, ficolin-H, CL-K1, PTX3). These lectins also come from multiple structural classes (Fig. 1C), including the C-type (calcium ion-dependent) lectins, pentraxins, and S-type (galectins) lectins.

The use of human lectins as glycan analysis probes has remained relatively limited due to technical challenges, including difficulties obtaining pure and natively-folded lectins (Vorup-Jensen 2014; Martínez-Alarcón et al. 2018) and their propensity to form oligomeric structures (Sheriff et al. 1994). Thus, our objective was to generate tagged recombinant human lectins tailored for use in multiple applications. We have established robust modular approaches, based on bacterial and mammalian cell expression systems emphasizing: 1. Production and purification methods that yield natively-folded, bioactive lectins by extraction of recombinant protein from soluble cellular fractions, rather than renaturation from inclusion bodies; 2. Strategies for the site-specific introduction of reporter tags; 3. Flexibility to substitute native oligomerization domains with alternative scaffolds encoding known oligomerization properties.

Figure 1A illustrates the standardized workflow for developing, validating, and applying GAPs to detect glycans in a complex community. The applications of the new GAPs include profiling co-existing microbial communities and microbial glycans, comparing the lectin binding specificities, and gaining comprehensive insights into lectin binding to other glycoproteins involved in host defense.

Production of GAPs from recombinant lectins

Commercially available plant and fungal lectins are used for imaging and labeling glycans (Rüdiger and Gabius 2001; Tsaneva and Van Damme 2020; Nazar et al. 2021). These are generally biotinylated or fluorophore-labeled using non-specific conjugation chemistry. We envisioned generating recombinant human lectins with small peptide tags of <10 amino acids that could be modified with different reporter groups.

Recombinant lectin production in bacterial cells

Escherichia coli-based expression was utilized to produce lectins that do not require glycosylation for folding and solubility (Table 2). We designed constructs to generate the carbohydrate recognition domains (CRDs) alone, or the CRDs with oligomerization neck domains, to preserve the native fold and binding valency. Additionally, for functionalization, the constructs were engineered with a small tag, an N-terminal Strep-II tag or a C-terminal sortase-mediated ligation (SML) LPXTG sequence. Using SML, the LPXTG motif allows facile and site-specific conjugation of affinity reporters (e.g. biotin) and spectroscopic labels (e.g. fluorophores) (Policarpo et al. 2014; Ghosh et al. 2023). The thermostability of purified GAPs was assessed using nano-differential scanning fluorimetry (nDSF) (Table 2). The GAPs produced using this approach facilitate targeted identification and enrichment of complex glycans from natural sources in various biochemical and cellular assays. Additionally, incorporation of a biotin tag allows multivalent presentation of lectin CRDs on the tetrameric streptavidin scaffold, mimicking avidity effects observed in physiological conditions.

Using the *E. coli* expression system, we produced five new GAPs. Three are based on galectins (S-type lectins) (Klyosov et al. 2008), an important family of human soluble lectins, which broadly recognize β-galactose-containing glycans and participate in diverse physiological and pathobiological roles (Vasta 2009). Unlike most human lectins, galectins are not glycosylated, allowing successful production in *E. coli*. The GAPs are based on CRDs from the chimeric galectin-3, and tandem-repeat galectins, including galectin-4 (Supporting information, Fig. S1A) and galectin-8 (Supporting information, Fig. S1B). Galectins have been produced previously as recombinant GST fusions (Poland et al. 2015; Paul et al. 2022; Dey et al. 2023), resulting in lectins that are presented as dimers. In contrast, our constructs have smaller tags, which can allow them to adopt native oligomerization states. These GAPs, engineered with a C-terminal His_{6x} tag for purification, and an N-terminal Strep-II tag for detection, enabled one-step affinity purification, yielding three functional lectin CRDs for characterization.

Building on our previous studies on the oral lectin zymogen granule protein ZG16B as a GAP for identifying cell-wall polysaccharides of specific oral commensals (Ghosh et al.

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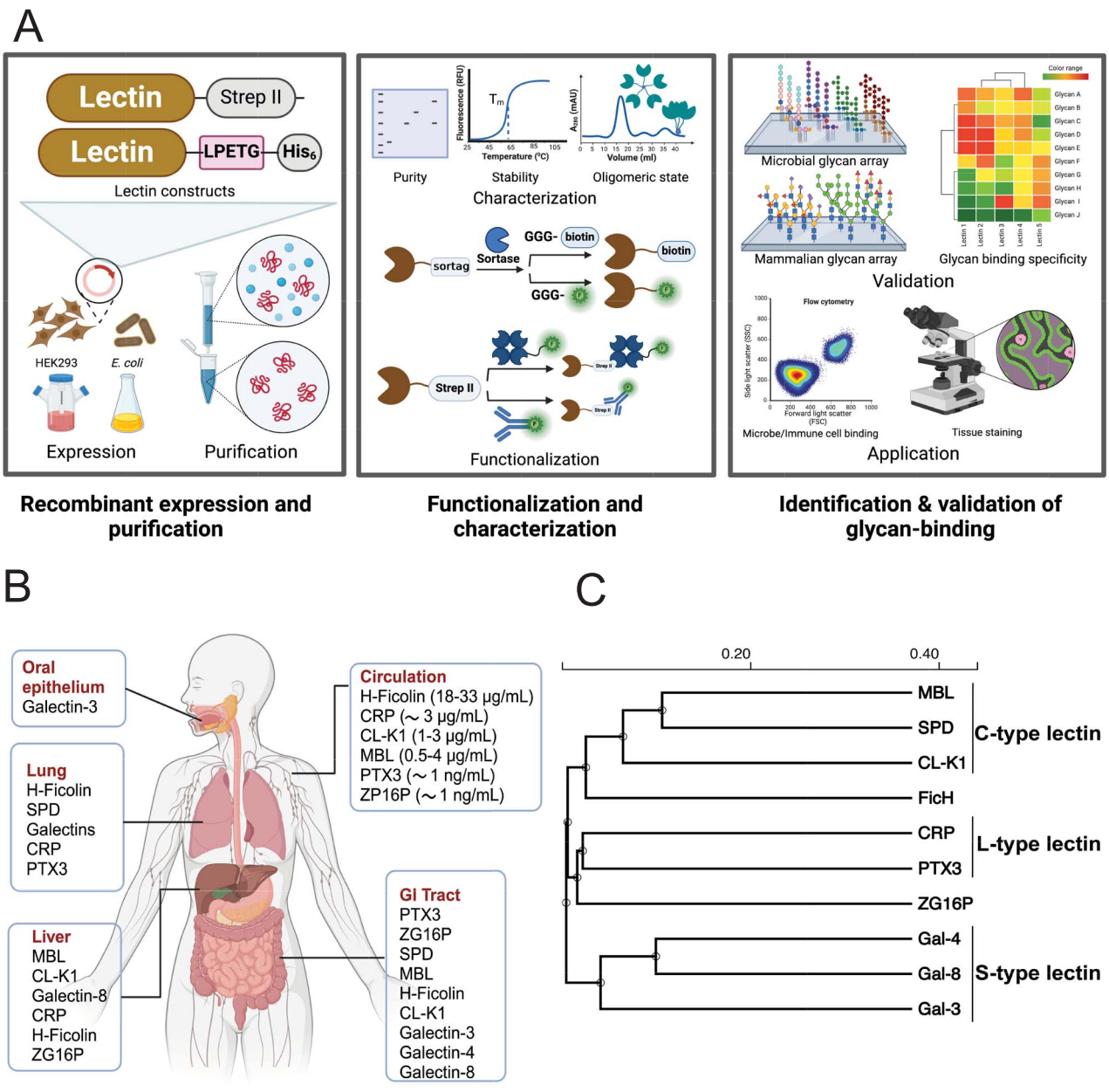


Fig. 1. Production of glycan analysis probes (GAPs) from human soluble lectins. A) Schematic diagram of the workflow of developing through recombinant production of human soluble lectins, their functionalization with reporter tags, and identification of their glycan binding partners. B) Tissue-specific distribution and abundance of human soluble lectins used in this study (adopted from (Peiffer et al. 2024). C) Guide tree representing the structural homology of the human soluble lectins used in the study, generated by multiple sequence alignment in Clustal omega. The scale indicates the length of the branches, indicating the evolutionary distance between the sequences.

2023), we also targeted the paralog intestinal lectin ZG16P (Kanagawa et al. 2011; Kumazawa-Inoue et al. 2011) by recombinantly producing the lectin in *E. coli*. The construct encoding the ZG16P-GAP excluded the signal sequence and contained both an additional C-terminal sortase tag (LPETG) and a His_{6x} tag (Supporting information, Fig. S1C). This construct was expressed, purified, and conjugated selectively with biotin (Supporting information, Fig. S1D). Its subsequent use in binding studies illustrates its utility (vide infra).

A GAP based on H-ficolin, one of the most abundant human ficolins in serum without significant functional annotation (Matsushita et al. 2002), was also developed. Recombinant production of ficolins in *E. coli* has been challenging as they require glycosylation (Garlatti et al. 2007; Hummelshoj et al. 2007) and disulfide bond formation (Ohashi and Erickson 2004) for folding, stability, and function. Overcoming these challenges, we successfully expressed an H-ficolin-GAP construct with the H-ficolin neck and fibrinogen-like domain (amino acid residues 81–299), and additional C-terminal sortase (LPETG) and His_{6x} tags in *E. coli*. The folded protein was purified from the soluble cellular fraction (Supporting information, Fig. S1E) and conjugated with biotin using SML. (Supporting information, Fig. S1F). The recombinant lectin oligomerized in solution as observed by the native gel electrophoresis (Supporting information, Fig. S1G) and dynamic light scattering (DLS) (Table 2, Supporting Information Fig. S2) (Garlatti et al. 2007). The H-ficolin-GAP also showed a folding transition

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Table 1. Biological significance and abundance of human soluble lectins chosen for GAP development in this study.

Lectin	Function	Abundance & Localization	References
Mannose Binding Lectin (MBL) (C-type)	Pattern recognition as in the first line of immune defense	Serum (0.3–4.0 μg/mL) and liver	(Arnold et al. 2006, Garred et al. 2016)
H-ficolin (FicH)	Activates lectin complement pathway	Serum (3–5 μg/mL) and lungs	(Matsushita et al. 2002, Endo et al. 2007, Gout et al. 2010, Cedzyński and Świerzko 2023)
Surfactant Protein D (SP-D) (C-type)	Exert host defense in lungs	Lung	(Kishore et al. 2006, Vieira et al. 2017)
Collectins (CL-K1) (C-type)	Binding to microorganisms	GI tract, lung, serum (1–3 μg/mL)	(Hansen et al. 2010, Ohtani et al. 2012, Selman and Hansen 2012, Henriksen et al. 2013)
Galectins (3,4,8) (Gal-3, -4, -8) (S-type)	Apoptosis, T-cell activation, autophagy-mediated antimicrobial activity	Lung, liver, GI tract	(Vasta 2009, Stowell et al. 2014, Wu et al. 2021, Wu et al. 2023, Jan et al. 2024)
C-reactive protein (CRP)	Activate complement pathway and facilitate antigen uptake	Serum (3 μg/mL)	(Gershov et al. 2000, Du Clos 2013)
Pentraxin 3 (PTX3)	Microbial defense, immunomodulation	Lung, GI tract, serum (1 ng/mL)	(Garlanda et al. 2002, Deban et al. 2011, Mantovani et al. 2013)
Zymogen granule protein (ZG16P)	Protein secretory, granule packaging	GI tract	(Kumazawa-Inoue et al. 2011, Bergström et al. 2016)

Table 2. Details of the glycan analysis probes (GAPs).

Heterologous expression	Lectin	Domain	M.W (kDa)	pI	E_{max} (M^{-1} cm^{-1})	Expression system	Solubility	Yield (mg/L)	T_m (° C)	Cumulant radius (nm)	Functional Tag
Mammalian	MBL	Full length	25.2	5.4	23,865	HEK 293 T	soluble	2	74.8 ± 0.04	6.9 ± 0.09	N-term Strep II
	ΔN-SP-D	Collagen-like domain+ +Neck +CRD	34	6.3	21,220	HEK 293 T	soluble	2	66.3 ± 0.1	40.8 ± 0.42	N-term Strep II
	CL-K1	Full length	27.6	5.2	25,815	HEK 293 T	soluble	2.5	45.3 ± 0.6[a]	17 ± 0.40	C-term Strep II
	PTX3	Full length	41.6	4.9	65,930	HEK 293 T	soluble		58.2 ± 0.01[b]	12 ± 0.05	C-term Strep II
	CRP	Full length	24.5	5.5	50,545	HEK 293 T	soluble	16	60.2 ± 0.01	4.5 ± 0.25	C-term Strep II
Bacterial	Gal3	CRD	16.63	9.8	9970	BL21(DE3)	soluble	18	59.8 ± 0.1	3.3 ± 0.45	N-term Strep II
	Gal4	Full length	38.36	9.3	42,860	BL21(DE3)	soluble	9	65.0 ± 0.1	3.5 ± 0.63	N-term Strep II
	Gal8	Full length	38.25	8.0	32,430	BL21(DE3)	soluble	12	53.2 ± 0.2	3.8 ± 0.52	N-term Strep II
	FicH	neck+ CRD	22.2	6.2	52,035	BL21(DE3)	Partially soluble	2.5	49.4 ± 0.6	5.2 ± 1.12	Biotin by SML[c]
	ZG16P	CRD	17.6	9.3	29,910	BL21(DE3)	soluble	6	54.6 ± 0.8	1.8 ± 0.32	Biotin by SML[c]

[a]CL-K1 showed a secondary T_m at 60.6 ± 0.6 °C. [b]PTX3 showed a secondary and tertiary T_m at 65.6 ± 0.4 and 70.5 ± 0.2 °C. [c]SML: sortase-mediated ligation method.

in nDSF (Supporting information, Fig. S3). Thus, despite the challenges of this protein class, the resulting multimer was sufficiently stable for biochemical studies.

We also targeted bacterial production of GAPs using the neck and CRD domains of several representative lectins from the C-type or collectin class (Holmskov et al. 2003) and pentraxin class (Deban et al. 2011). However, in these cases, bacterial expression failed to yield a protein with a functional oligomeric fold, possibly due to non-native disulfide bond formation (Kishore et al. 2006; Inforzato et al. 2008) and the lack of glycosylation on the N-terminal accessory domains, which influences stability and solubility (Tsuji et al. 2001; Inforzato et al. 2006; Garred et al. 2016; Wesener et al. 2017). To overcome these problems, a eukaryotic expression system was used for a subset of lectins that relied on posttranslational modification for proper folding and function.

Recombinant lectin production in mammalian cells

For the development of GAPs based on pentraxins and collectins (Supporting information, Fig. S4), lectins were expressed in HEK-derived mammalian cells (Table 2). The collectin constructs were designed based on guidance from the literature and predicted structures (Drickamer 1993; Sheriff et al. 1994; Weis and Drickamer 1994; Drickamer 1999). All

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lectins were full length with the exception of SP-D, whose N-terminus was truncated to avoid aggregation (ΔN-SP-D) The lectins were engineered for secretion to facilitate isolation (Wesener et al. 2017). An N-terminal Strep-II tag was added to the constructs for direct antibody-based or streptactin-mediated detection. This tag also facilitated oligomerization using streptavidin. The formation of higher-order oligomers of the full-length recombinant lectins was supported by dynamic light scattering (DLS) measurements (Table 2, Supporting information, Fig. S5) and western blot analysis under non-reducing conditions. The stability of the recombinant lectins in solution was assessed using nDSF (Table 2, Supporting information, Fig. S6).

We focused on three members of the human soluble collectin family (Håkansson and Reid 2000): mannose-binding lectin (MBL) and collectin kidney-1 (CL-K1), which are predominantly found in blood, and pulmonary lectin surfactant-associated protein D (SP-D), which is found in the lung and gut. These are all representative effector proteins of the innate immune system (Casals et al. 2019; Cedzyński and Świerzko 2023). The representative N-terminal Strep-II tagged GAPs (Supporting information, Fig. S4A–C) from these collectins varied in their oligomeric state as observed in DLS (Table 2, Supporting information, Fig. S5), DSF (Table 2, Supporting information, Fig. S6), and gel electrophoresis analysis (Supporting information, Fig. S4A–C).

Another essential class of lectins, critical in the complement pathway-mediated immune response, are the pentraxins (Inforzato et al. 2006; Deban et al. 2011). These proteins are abundant in the blood, enhance opsonophagocytosis (Mold et al. 2001), and tune inflammation (Gershov et al. 2000; Du Clos 2013). We sought to evaluate their binding interactions as the ligand specificity of these lectins is not annotated. We chose the full-length short pentraxin C-reactive protein (CRP) (Supporting information, Fig. S4D) and long pentraxin (PTX3) (Supporting information, Fig. S4E) as representative L-type lectins for GAP development. Both CRP and PTX3-GAP formed multimers in solution as assessed by non-reducing gel electrophoresis (Supporting information, Fig. S4D and E) and DLS analysis (Table 2, Supporting information, Fig. S5).

Successful production of GAPs that adopt native-like conformations and oligomerization states enables the investigation of cognate, biologically relevant microbial glycan targets. Additionally, using the GAPs in various experimental formats (e.g. arrays, mucin-binding, and tissue staining) can reveal previously unrecognized glycosylation patterns, identify knowledge gaps in our understanding of glycobiology, and identify potential biomarkers for various conditions.

Glycan-binding specificity of GAPs

GAP analysis with bacterial glycan microarrays

Glycan arrays can reveal range of potential glycan ligands with limited protein requirements and represent the most high-throughput method for identifying glycan-binding specificity. For the array analysis, we deployed both microbial and human glycan arrays. To investigate the ligand specificity of the functionalized lectin CRDs, we first used a custom bacterial glycan array from the Scripps Research Institute Microarray Core Facility, printed with distinct polysaccharides isolated from a range of microbes (Stowell et al. 2014). The bacterial glycan array includes over 300 microbial polysaccharides printed on an amine-reactive N-hydroxysuccinimidyl ester (NHS)-activated glass slide in a 4∗12 grid matrix, with each grid having six replicates of each glycan. The array features glycans primarily isolated from gram-negative bacteria belonging to the order *Enterobacteriales* (Stowell et al. 2014), including the genuses *Escherichia, Klebsiella, Proteus, Providencia, Salmonella, Shigella,* and *Yersinia*. Capsular polysaccharides derived from the gram-positive bacteria *Streptococcus pneumoniae* are also included in the microarray. The printed polysaccharides are enriched in non-mammalian monosaccharides, such as FucNAc, D-FucNAc, ManNAc, 3′-deoxy sugars (KO and KDO), and heptoses. A similar bacterial glycan array has previously been used to predict and validate host–microbe interactions. For example, the binding of galectin-8 to immobilized glycan epitopes derived from the gram-positive microbe *S. pneumoniae* predicted strain-specific binding and antimicrobial activity against whole microbial cells, supporting that the array could provide insight into physiologically relevant glycan-binding properties (Wu et al. 2023). Despite these studies and the importance of innate immune lectins for bacterial detection, few human lectins have been probed on this array.

We evaluated the binding of soluble human lectins from the GAP production platform outlined previously on the bacterial glycan array. These included the S-type lectins (galectin-3, -4, -8), pentraxins (CRP, PTX3), collectins (MBL, ΔN-SP-D, CL-K1), as well as H-ficolin and ZG16P. The binding intensity of the GAPs to each glycan sample was calculated as an average of integrated spot intensities from six replicates. Internal grid controls, including dye samples and printed null spots, aided in validating the consistency of the assay results.

Due to the lot-to-lot variability in the construction of arrays and fluorescence readings across experiments, we normalized binding data for each lectin such that the highest binding value above our binding signal threshold was set to 100%, and binding to every glycan was plotted as a percentage of binding relative to this value (Fig. 2A, Supporting information, Spreadsheet 1). We set the top 10% of the highest normalized fluorescence intensity (100%) as the threshold to identify binders of GAPs on the bacterial glycan array (Supporting information, Spreadsheet 1) (Cao et al. 2019). This approach enabled us to independently determine the binders for each GAP, even when the binding strength is relatively weak. Additionally, the total fluorescence binding intensity is considered to assess the binding profile of GAPs for one-to-one comparison (Fig. 2B, Spreadsheet 2). We note, however, that intensity is not a direct measure of affinity but only a means by which to identify putative protein-glycan interactions.

We observed distinct binding profiles for the GAPs on the bacterial glycan arrays (Fig. 2A and B, Supporting information, Fig. S7), supporting the specificity of the GAP-glycan interactions. Several GAPs, including MBL, ΔN-SP-D, CRP, and galectins-3, -4, and -8 exhibited more readily apparent binding to multiple microbial glycans compared to other GAPs (Fig. 2B, Supporting information, Spreadsheet 2). We also noted that the C-type lectins MBL and SP-D preferred similar microbial glycans, including the lipopolysaccharides (LPS) from pathogenic strains of gram-negative *Yersinia pestis* containing heptose KO (D-glycero-D-talo-oct-2-ulosonic acid) and KDO (3-deoxy-D-manno-oct-2-ulosonic acid) (Supporting information, Fig. S7, Spreadsheet 1, 3). These lectins also showed a broad range of binding to O-specific polysaccharides (OPS) or O-antigens from other

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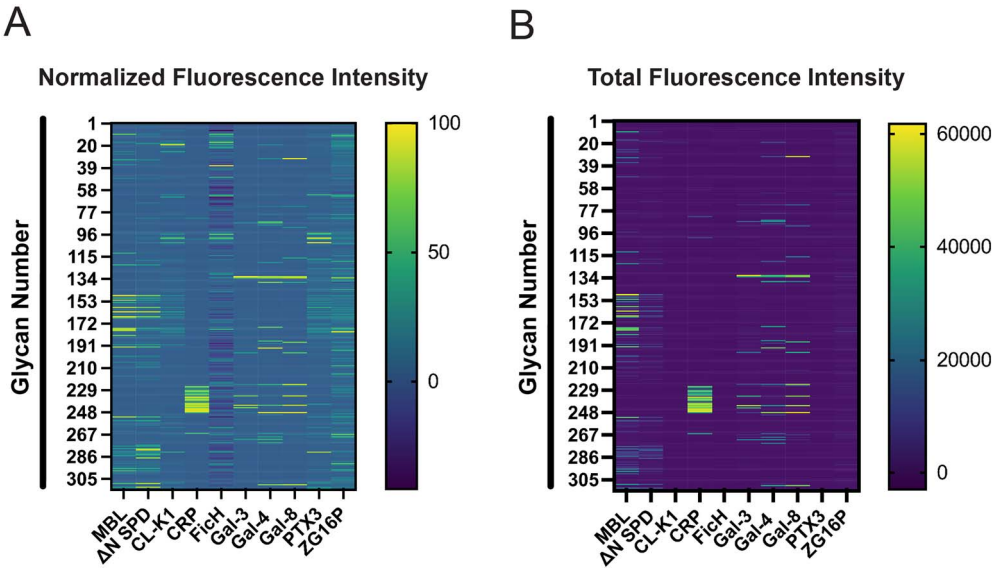


Fig. 2. Microbial glycan binding specificity of GAPs: Glycan binding profile of GAPs on custom bacterial glycan array (from TSRI MicroArray Core Facility, the Scripps research institute) represented as heatmaps. A) Heatmap generated based on the normalized values of relative fluorescence intensity for each lectin considering the highest binding value as 100% binding (on the scale). Data are presented as the mean ± s.d. ($n = 6$ technical replicates). B) Heatmap generated based on the total binding intensity quantified as relative fluorescence unit (on the scale).

pathogenic strains of *Shigella flexneri*, *E. coli,* and *Klebsiella pneumoniae* (Supporting information, Fig. S7, Spreadsheet 1). In contrast, CRP, which is typically used as biomarker of general inflammation (Sproston and Ashworth 2018), specifically interacts with capsular polysaccharides (CPS) from *S. pneumoniae*. Galectins-3, -4, and -8 interacted with a diverse set of glycans but also shared similarities in their binding profiles, showing a strong preference for OPS and LPS containing colitose, D-glucofuranose, and D-hexose (Supporting information, Fig. S7, Spreadsheet 1, 3).

Despite our success in identifying glycan targets for the lectins above, a notable limitation of the bacterial glycan array is that it is generated using a limited set of glycans isolated from natural resources and lacks glycans isolated from species other than pathogenic bacteria. Importantly, it does not yet encompass the diversity of monosaccharide building blocks or glycoside linkages of the constituent glycans found in microbial glycomes that can be achievable through combined synthetic strategies and extraction from natural resources (Geissner et al. 2019). GAPs developed from CL-K1, H-ficolin, PTX3, and ZG16P exhibited weaker signals, likely due to the absence of their cognate targets on the bacterial glycan array (Fig. 2B). This result highlights the need for screening the GAPs on a more diverse glycan array. Despite the lack of diversity, certain GAPs with weak microbial glycan binding revealed intriguing profiles in the normalized analysis (Fig. 2A). For example, the long pentraxin PTX3, although sharing structural similarities at the CRD with CRP, showed a different binding profile; it bound to a broad range of glycans with a preference for OPS from *Proteus mirabillis* and *S. flexneri* containing ethanolamine-modified sugars (Supporting information, Fig. S7, Spreadsheet 1, 3) (Toukach et al. 2001). H-ficolin mainly binds to OPS from different O-serotypes of *P. mirabillis* containing a D-galacturonic or D-glucuronic acid, often attached to an L-amino acid via an amide bond (Supporting information, Fig. S7, Spreadsheets 1 and 3) (Arbatsky et al. 1999).

The monomeric lectin ZG16P bound a broad range of microbial glycans, although we speculate that many may be non-specific. Previously, ZG16P was reported to bind to several gram-positive bacteria, including *Lactobacillus jensenii and Enterococcus faecalis* (Bergström et al. 2016). The oral lectin ZG16B, a paralog of ZG16P, is observed to prominently bind to a set of oral gram-positive bacteria, including *Streptococcus vestibularis* (Ghosh et al. 2023). Therefore, with a limited set of glycans from gram-positive bacteria on the bacterial glycan array, it may not be feasible to identify physiologically relevant glycan targets of ZG16P without further study.

GAP analysis with mammalian glycan microarrays

To complement the bacterial glycan array analysis, we used the commercially available mammalian glycan array 300 (Ray Biotech). The Ray Biotech array, featuring a subset of glycans on the mammalian array available through the Center for Functional Glycomics (CFG), includes 300 mammalian glycans comprising N-glycans, glycolipid glycans, human milk oligosaccharides and sialic acid-rich tandem epitopes. Although the glycan array provides sufficient diversity for determining lectin binding, the limited availability of O-linked glycans and glycosaminoglycans (GAGs) makes it difficult to completely rule out the possibility that some GAPs may bind to both host glycans and microbial targets. Indeed, we have observed simultaneous binding to both highly glycosylated mucins and bacteria (Ghosh et al. 2023).

In general, the GAPs tested exhibited a weaker affinity for mammalian glycans compared to microbial glycans (Supporting information Fig. S8, Spreadsheet 4, 5). This selectivity highlights the propensity of soluble human lectins to bind microbial glycans. However, various GAPs, particularly those based on galectins, demonstrated affinity for selected mammalian glycans as expected from the data available from the CFG website (Supporting information Fig. S8, Spreadsheet 4, 5) (Citation: National Center for Functional Glycomics, Legacy CFG Microarray Data; https://research.bidmc.org/ncfg/microarray-data/legacy-cfg-microarray-data; Accessed 4/25/25). This observation may result from the lectins influencing microbial colonization by binding to the host glycoproteins, especially host mucins, or by

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forming ternary complexes that associate host and microbe (Bergström et al. 2016; Ghosh et al. 2023).

GAP binding to human host mucins

We also explored GAP binding to other host glycans found in physiologically relevant niches. Our objective was to provide insights into the role of lectins in facilitating host–microbe interactions. The mucosal barrier is central to regulatory activities in host-microorganism interactions, including receptor-mediated endocytosis, cellular recognition, and microbial adhesion (Jeffers et al. 2010; Frenkel and Ribbeck 2015). As a component of mucosal immunity, many lectins interact with the highly glycosylated mucins, which are glycoproteins with O-linked glycans in the mucus layer (Chairatana and Nolan 2017; Peiffer et al. 2024). Mucins can modulate inflammatory responses by forming mucus layers on epithelial cell surfaces that protect tissues from bacterial invasion (Frenkel and Ribbeck 2015). The crosstalk of soluble lectins and mucins often influences microbial colonization and the bacterial growth regulatory effects of the mucosal barrier.

A number of studies have identified lectin-mucin interactions highlighting their potential functional consequences. Galectin-3 has been shown to bind the gastrointestinal mucin MUC2 and respiratory tract mucins MUC5AC and MUC5B (Diehl et al. 2024) when regulating innate immune properties of airway mucosal surface; ZG16P prevents bacterial aggregation on colon epithelial cells by working together with the colon mucin MUC2 (Bergström et al. 2016); the oral lectin ZG16B recruits the oral mucin MUC7 on opportunistic pathogens in the oral cavity and maintains microbial homeostasis (Ghosh et al. 2023). However, assessing the relevance of lectin-mucin interactions is challenging due to the high complexity of O-glycosylated mucins and the multivalent, cooperative nature of the interactions. The GAPs represent physiologically relevant lectin-based analytical tools to address these challenges. We employed biotinylated or Strep-II-conjugated GAPs to capture the glycan-specific interaction between mammalian mucins and human soluble lectins. GAPs can be used to identify specific glycan epitopes on the mucins that can modulate the mucosal barrier properties, influencing microbial adhesion, colonization, and potentially pathogenicity. Additionally, the biotinylated GAPs can also be used to create the multivalent display required for these interactions, mimicking aspects critical for physiological binding.

GAPs produced from *E. coli* were screened for their interaction with crude or extracted porcine mucins MUC2, MUC5AC and MUC5B by a dot-blot assay. The relative binding of GAPs to different mucins was normalized to the highest intensity integrated from the dot blot signals. GAPs from galectin-4 (Gal-4) and galectin−8 (Gal-8) bound strongly to MUC2, MUC5AC and MUC5B (Fig. 3A and B, Supporting information, Fig. S9A and B). The binding of Gal-8 was inhibited by the addition of 50 mM lactose as a competitive inhibitor, indicating that the interaction is mediated through mucin glycan recognition (Carlsson et al. 2007). However, the interaction of Gal-4 with the mucins was not disrupted by lactose, likely due to its stronger affinity for the N-acetyllactosamine (LacNAc) repeats present in the colonic mucins relative to lactose, as shown previously (Holmén Larsson et al. 2009; Elzinga et al. 2024).

The ability of GAPs to uniquely bind different mammalian mucins was assessed for ZG16P and H-ficolin. As these lectins are not yet known to bind mucin glycans, we initially investigated them with commercially available porcine Type-II and Type-III mucins. ZG16P-GAP showed binding to the crude porcine mucins (Fig. 3C, Supporting information, Fig. S9C). The lectin binding was reduced when mucins were treated with 1 mM periodate, suggesting that mucin O-glycans mediated the binding. When screened for binding to purified mucins, ZG16P showed binding to MUC2, MUC5AC, and MUC5B in a concentration-dependent manner (Fig. 3D, Supporting information, Fig. S9D). In contrast, the GAP from H-ficolin did not show glycan-dependent binding to crude porcine mucins (Supporting information, Fig. S10A and B) or a concentration-dependent binding to the purified porcine mucins (Supporting information, Fig. S10C and D). These differences in mucin binding are consistent with the localization of the lectins: H-ficolin is typically found in the blood, while ZG16P is present in the gut. Thus, only the latter lectin resides at a mucosal barrier. Similar reasoning can be applied to rationalize the binding difference between the galectins and ZG16P. Galectins showed stronger binding to airway pathway mucins (MUC5AC and MUC5B), while the latter showed a preference for the intestinal mucin (MUC2).

The O-glycan structures in MUC2, MUC5AC, and MUC5B differ in the overall complexity and length of their glycan chains. MUC2 (Thomsson et al. 2012) typically has shorter, less extended O-glycans than the longer and more complex structures found on MUC5AC and MUC5B (Holmén et al. 2004). Additionally, MUC5AC core-2 glycans are often elongated with lactosamine (Gal-GlcNAc) units, and have fewer sialylated structures compared to MUC2, making the former mucin a better target for galectins (Holmén Larsson et al. 2013). These differences in glycosylation pattern can contribute to the distinct and varied mucin-binding capabilities of GAPs and underscore the specialized roles of different lectins in mucin recognition and interaction.

Application of GAPs in tissue analysis

The GAPs can also serve as tools for tissue staining, as altered glycosylation of secretory glycoproteins in the mucus layer is common in various colonic and respiratory conditions (Chatterjee et al. 2020; Grondin et al. 2020). Therefore, GAPs could be useful non-invasive markers for differentiating diseased from healthy tissues. We therefore assessed the utility of these reagents for staining tissues.

We used a FITC-conjugated galectin-3 (Gal-3) GAP for fluorescent lectin histochemistry (Fig. 4). Fixed murine ileum and colon tissues were sectioned, mounted on a glass slide, and incubated with FITC-conjugated Gal-3 GAP. The Gal-3 GAP readily stained the extracellular matrices of mucus-producing goblet cells in murine intestinal tissue. The goblet cells form the inner mucus layer rich in MUC2 that separates the colonizing bacteria in the outer mucus membrane from the epithelium (Kim and Ho 2010; Pelaseyed et al. 2014). Mice with a disrupted inner mucus layer, lacking mucin MUC2, showed symptoms of colitis due to penetration of bacteria from the outer to the inner mucus layer (Van der Sluis et al. 2006; Johansson et al. 2008). These observations suggest that the Gal-3 GAP, likely recognizing the mucin glycans (Diehl et al. 2024) in the inner mucus scaffold on goblet cells, can effectively reveal the organization of the intestinal tissues.

Application of GAPs in identifying specific microbial populations in a complex microbiome

An advantage of soluble GAPs is that they can be used to assess the preferences of lectins for binding to microbes within

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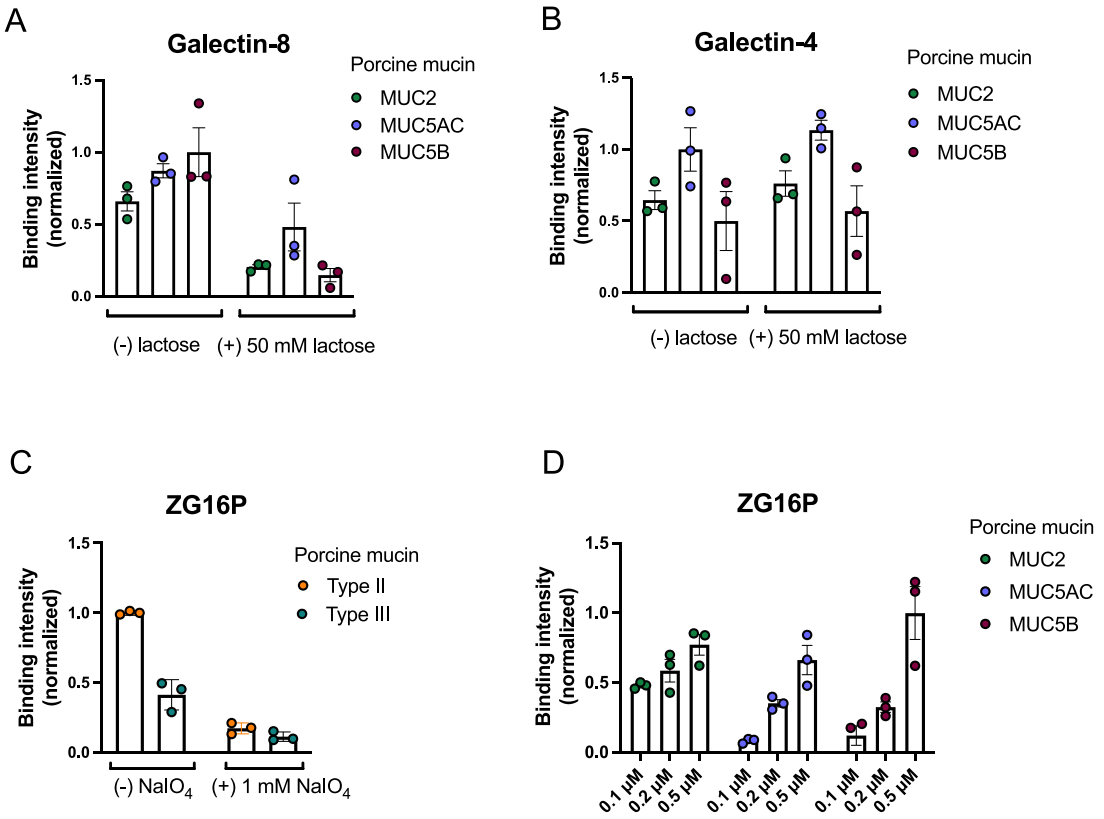


Fig. 3. Mucin binding of GAPs: Binding profile of A) Galectin-8 GAP and B) Galectin-4 GAP with purified porcine mucins MUC2, MUC5AC, and MUC5B. The binding intensity was quantified from the mucin dot blots probed with strep-II tagged GAPs at 250 nM concentration, followed by washing with mucin binding buffer with or without 50 mM lactose as a competitive inhibitor. C) Binding of ZG16P-biotin GAPs at a 100 nM concentration to crude porcine mucins, treated with 1 mM sodium periodate to oxidize the vicinal-diols of mucin O-glycans. Data are represented as mean ± SEM. (n = 3 technical replicates). D) Binding of ZG16P-biotin GAP to purified porcine mucins MUC2, MUC5AC, and MUC5B at different concentrations. Data are represented as mean ± SEM. (n = 3 technical replicates).

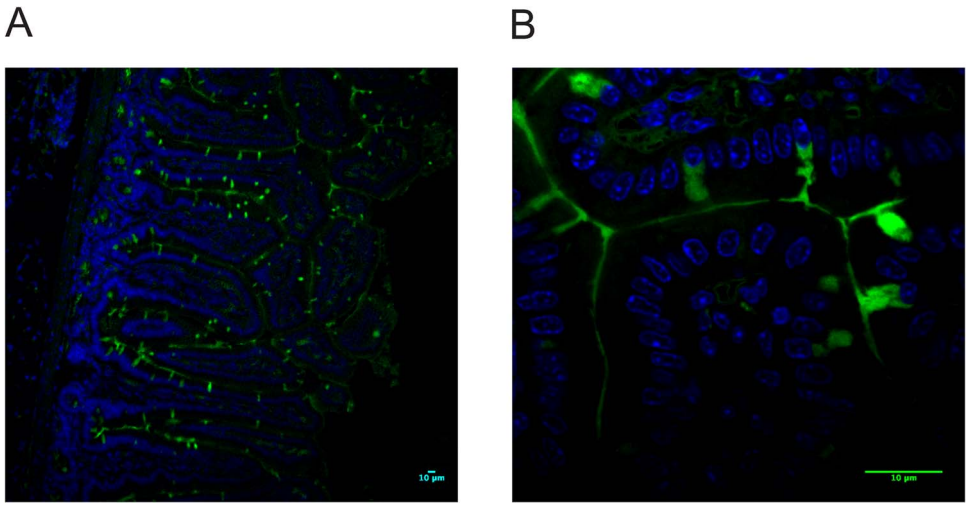


Fig. 4. Staining of the extracellular matrices of mucus-producing goblet cells in murine intestinal tissue by FITC-conjugated galectin-3 GAP. Surgical murine ileum and colon tissue were probed with galectin-3-FITC (shown in green) and visualized with the RPI spinning disc confocal microscope at A) 10× and B) 63× objectives. The GAP stained the inner mucus layer produced by the goblet cells. Nuclei are stained with DAPI (blue).

complex communities. This issue is relevant, as we showed previously that lectins can exhibit species preferences within a community that are not apparent when examining binding to single species (McPherson et al. 2023). To assess the utility of GAPs in profiling lectin-microbe interactions in native microbial communities, we screened seven representative GAPs for their binding to fecal bacteria samples collected from six healthy individuals and six inflammatory bowel disease (IBD) patient donors (Figs 5–7). Microbes in stool homogenates from these twelve donors were stained with GAPs derived from zymogen granule proteins, collectins and pentraxins. Fluorescence-activated cell sorting (FACS) was employed to isolate the lectin-bound and unbound bacterial populations.

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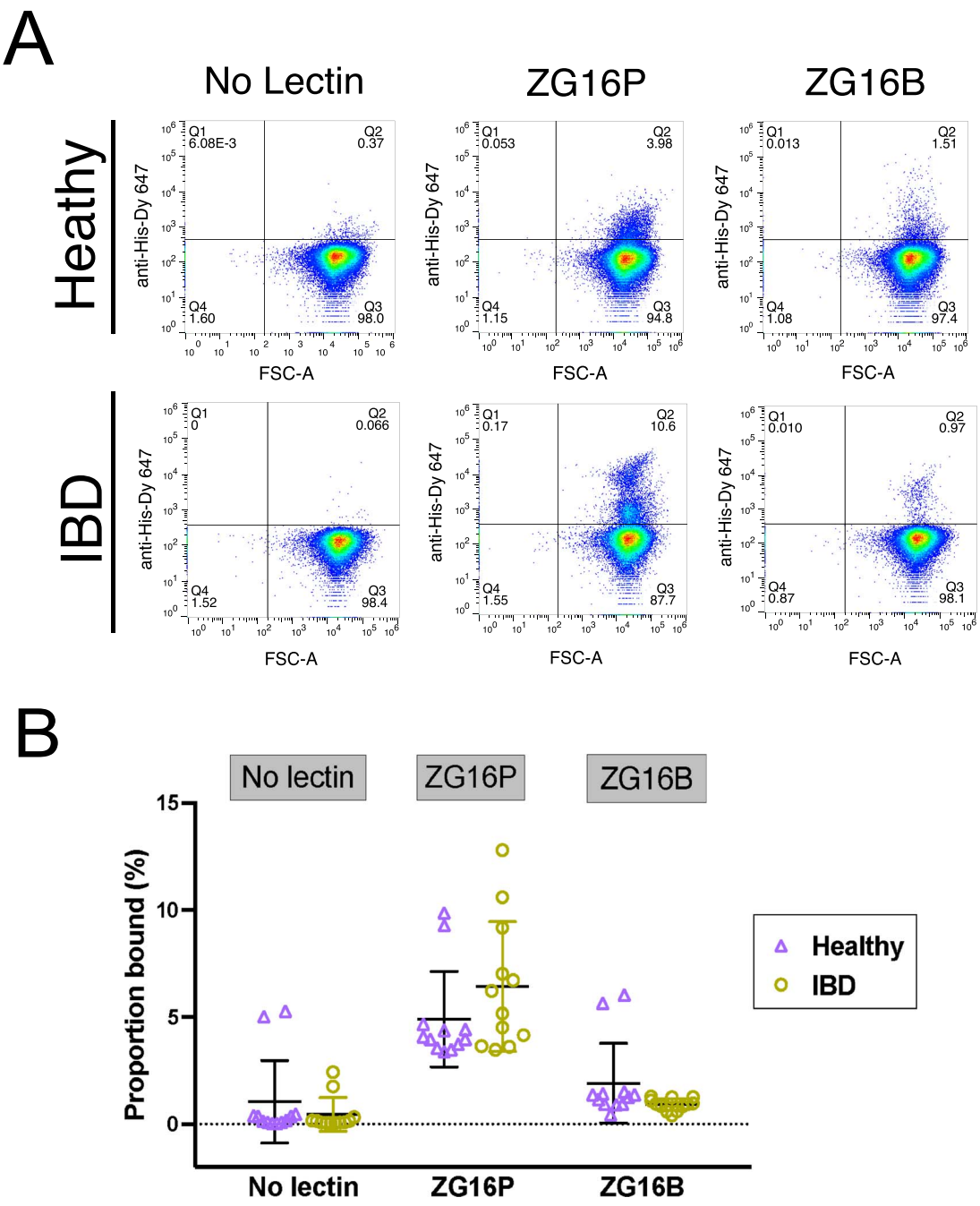


Fig. 5. Flow cytometry analysis of human stool microbiome from 6 healthy and 6 IBD donors for differential binding of GAPs from zymogen granule proteins ZG16P and ZG16B. A) Scatter plots showing microbial fraction from a representative healthy (top row) and an IBD patient donor (bottom row) as FSC-A against lectin binding (anti-His DY649). Panels show no lectin control (column 1), and His-tagged ZG16P (panel 2), and ZG16B (panel 3) binding to the fecal microbial samples. B) Dot plots representing lectin-binding data from 6 healthy and 6 IBD bacteria samples (n = 6), the horizontal and vertical line showing the mean and the SEM, respectively) proportion bound is calculated by determining the percentage of cells that fall in the His-pos/FSC-pos fraction (Q2).

We observed distinct binding preferences of seven GAPs for fecal bacterial samples. One of the most interesting comparisons is between the GAPs derived from the intestinal lectin and ZG16P and the oral lectin ZG16B. We previously used ZG16B in dental swab microbiome profiling (Ghosh et al. 2023), where ZG16B bound to specific oral microbes that ZG16P did not recognize. In this study, we found that ZG16P-GAP exhibited stronger binding to fecal samples than did the ZG16B-GAP. The former exhibited robust binding to fecal bacteria in both healthy and IBD samples (Fig. 5A and B), while the oral lectin showed little binding. These data underscore the importance of tissue-specific lectin expression for lectin-glycan interactions.

Among the collectins, MBL, ΔN-SP-D, and CL-K1 all bound to fecal bacteria. Interestingly, MBL and ΔN-SP-D showed similar calcium-dependent binding patterns, interacting with subsets of bacteria in healthy and diseased samples (Fig. 6A and B). In contrast, CL-K1 exhibited a

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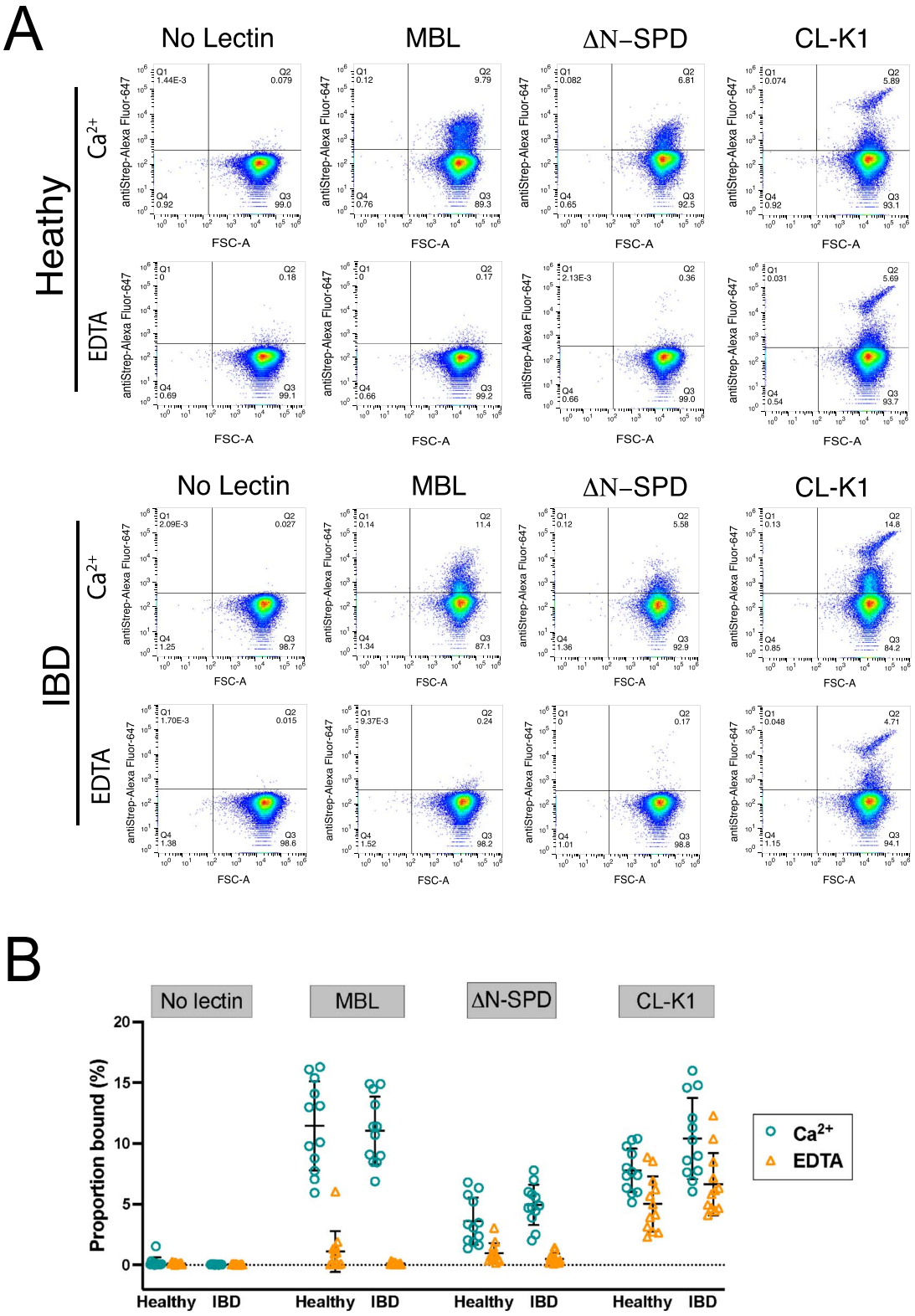


Fig. 6. Flow cytometry analysis of human stool microbiome from 6 healthy and 6 IBD donors for differential binding of GAPs from collectins MBL, ΔN-SP-D and CL-K1. A) Scatter plots showing microbial fraction from a representative healthy (top panel) and an IBD patient donor (bottom panel) as FSC-A against lectin binding (anti-strep Alexa Fluor 647) in the presence of Ca^{2+} (top row, each panel), or EDTA (bottom row, each panel). Panels show no lectin control (column 1), and anti-strep MBL (panel 2), ΔN-SP-D (panel 3) and CL-K1 (panel 4) binding to the fecal microbial samples. B) Dot plots representing lectin-binding data from 6 healthy and 6 IBD bacteria samples (n = 6), the horizontal and vertical line showing the mean and the SEM, respectively. Proportion bound is calculated by determining the percentage of cells that fall in the strep-pos/FSC-pos fraction (Q2).

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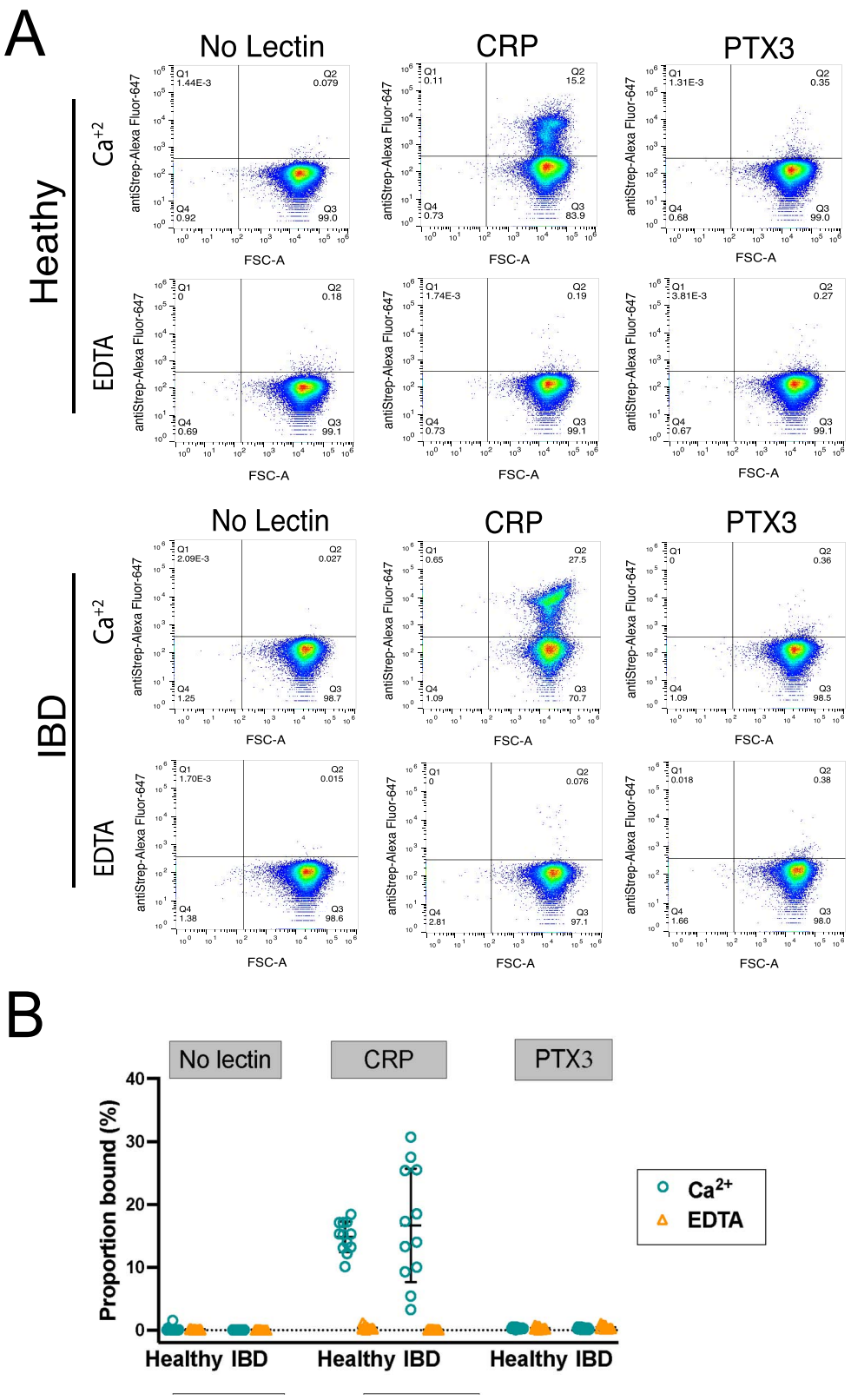


Fig. 7. Flow cytometry analysis of human stool microbiome from 6 healthy and 6 IBD donors for differential binding of GAPs from pentraxins CRP and PTX3. A) Scatter plots showing microbial fraction from a representative healthy (top panel) and an IBD patient donor (bottom panel) as FSC-A against lectin binding (anti-His DY649) in the presence of Ca^{2+} (top row, each panel), or EDTA (bottom row, each panel). Panels show no lectin control (column 1), and anti-strep CRP (panel 2), and PTX3 (panel 3) binding to the fecal microbial samples. B) Dot plots representing lectin-binding data from 6 healthy and 6 IBD bacteria samples ($n = 6$), the horizontal and vertical line showing the mean and the SEM, respectively) proportion bound is calculated by determining the percentage of cells that fall in the strep-pos/FSC-pos fraction (Q2).

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higher binding preference for fecal bacteria from IBD patients, binding 2–3% more bacteria in IBD samples than in healthy controls (Fig. 6A and B).

The pentraxins have been studied for their abilities to recognize pathogens and promote complement deposition and phagocytic uptake (Bottazzi et al. 2010; Foo et al. 2015). Less is known about their ability to interact with components of the microbiome. No detectable binding of PTX3 to fecal bacteria was observed (Fig. 7A and B). In contrast, CRP bound to a distinct subset of fecal bacteria in healthy and IBD samples, engaging about 20% of the population. These results highlight the specificity of GAPs in recognizing bacterial species within native communities. They also suggest the potential for identifying microbial signatures associated with health and disease.

Discussion

A significant current challenge in glycobiology is elucidating the details of microbial cell envelope glycoconjugates that mediate host attachment and drive pathogenesis. Plant and fungal lectins (Rüdiger and Gabius 2001; Tsaneva and Van Damme 2020) have been widely used to characterize mammalian glycans. However, due to the greatly expanded diversity of bacterial glycomes (Imperiali 2019), complementary tools are needed to reliably detect and characterize physiologically relevant glycan epitopes on the microbial glycocalyx and effectively distinguish them from host glycans. Human lectins, with their capacity to recognize distinct glycan epitopes, especially within diverse and complex microbiomes, offer a potential solution to the limitations of current lectin-based approaches. Human lectins, as a part of the innate immune system, have evolved to distinguish between host and microbial glycosylation. Motivated by this, we developed a strategy to generate glycan analysis probes (GAPs) from human soluble lectins. The result is a toolset for advancing our understanding of the precise protein-glycan interactions in host-microbe communication.

The development of GAPs based on human lectins will enhance our understanding of the characteristics of human soluble lectins across different structural classes, tissue-specific expression, roles in immune responses, and adaptive divergence in molecular recognition for diverse glycans. The presented experimental strategy is also generalizable, making it amenable to expansion to other lectins or glycan-binding proteins. Furthermore, we demonstrate that the engineered lectin-based probes are isolated in their correctly folded and relevant oligomeric states. The ability to produce natively folded GAPs and precise functionalization for glycan detection marks a practical advancement in studying complex biological interactions based on bacterial glycoconjugates. Critically, GAPs offer a new platform for identifying pathogens and defining clinically significant glycan epitopes for antibiotic development.

An advantage of using GAPs in decoding host–microbe interactions is the potential to perform unbiased searches for ligand specificity at a high throughput scale. For example, GAPs developed from human lectins ZG16B, MBL, and Intelectin 1 (IntL-1) have been used to recognize and isolate the microbial targets of the lectin from complex microbiota (Ghosh et al. 2023; McPherson et al. 2023). Furthermore, the GAP from oral lectin ZG16B could identify cell wall polysaccharides on the surface of oral commensals that mediate specific lectin-microbe interactions (Ghosh et al. 2023). Therefore, GAP applications provide top-down approaches for glycan identification, facilitating analysis of glycan binding across different species and in unfractionated microbiome samples.

GAPs offer a glycan profiling approach that complements the recently developed human lectin-based arrays (Benjamin et al. 2024). These arrays have proven crucial in high throughput screening of lectin interactions with isolated microbial and fungal polysaccharides, as well as viral glycoproteins, revealing unique binding targets of the lectins. Both approaches share some lectin representatives, such as MBL, SP-D and the galectins, but provide distinct insights: while human lectin arrays reveal the overall glycosylation profile of a sample, GAPs identify glycan preferences in the context of cells and heterogeneous samples. Therefore, by combining GAPs with human lectin arrays, researchers can achieve a comprehensive understanding of lectin specificity across a wide range of microbial and mammalian glycans in complex systems.

In this study, we deployed the GAPs on bacterial glycan arrays to identify specific microbial targets from a bottom-up perspective. Additionally, we used them on mammalian glycan arrays to understand the cross-reactivity of human lectins across diverse glycomes. Glycan array screening is critical to ensure that the glycosylation of the GAPs does not confound recombinant GAP recognition. This issue is especially relevant for GAPs produced in mammalian cell culture, as they are secreted proteins and, therefore, glycosylated. Despite these potential concerns, we observed that the selectivity of the different GAPs on glycan arrays indicates that their binding is driven by recognition of the immobilized glycans and not merely by protein glycosylation. Additionally, screening GAPs on bacterial and mammalian glycan arrays enabled an extensive cross-reactivity assessment, a critical consideration in the contexts of host–microbe interactions.

The custom bacterial glycan array used in this study features more than 300 naturally isolated microbial glycans comprising O-antigens from pathogenic gram-negative bacteria and cellular polysaccharides from selected gram-positive bacteria. Given the diversity of microbial glycans (Imperiali 2019), even these valuable arrays can only represent a small fraction of the available glycans. However, despite the relatively low diversity score of the custom bacterial glycan array relative to other reported arrays (Geissner et al. 2019), several GAPs still revealed potential binding candidates, underscoring the effectiveness of the probes in targeting specific glycans even in less diverse environments. More diverse glycan arrays, comprising glycans from commensal and pathogenic bacteria, will be very valuable for determining the ligand binding specificity of other GAPs and, importantly, for identifying differentiating characteristics between commensals and pathogens.

A strength of the GAP platform is its utility in analyzing cell binding specificity within diverse communities. For example, flow cytometry-based sorting offers a top-down approach for identifying glycan-mediated host–microbe interactions within complex microbial communities. When applied to the fecal samples, which reflect the vast diversity of the gut microbiome, GAPs could identify specific subsets of bacteria, presumably mediated through the microbial glycoconjugates.

Converting the paralogous lectins ZG16B and ZG16P into GAPs allowed a comparison of their specificity for microbes. We had previously observed that ZG16B, an abundant

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salivary lectin, interacts with commensal oral bacteria, including *S. vestibularis* (Ghosh et al. 2023). In contrast, the gut lectin ZG16P had no detectable interactions with oral bacteria. We also found that ZG16P binds robustly to fecal bacteria, while ZG16B shows limited interactions with these communities. These data indicate a link between the physiological location of a lectin and its microbial partners, highlighting the importance of lectin binding to distinct microbial populations.

The composition of gut microbiota changes in a diseased situation, e.g. inflammatory bowel disease (Vich Vila et al. 2018), leading to the disruption of the regular lectin-glycan interaction. The differential binding profiles of GAPs to gut microbes in health and disease conditions suggest their utility in mapping disease-associated microbial signatures. Furthermore, GAPs can enable the identification of glycan epitopes on bacteria responsible for colonization in diseased states, shedding light on how glycan-mediated interactions shift under pathological conditions. These opportunities underscore GAPs for lectin-based microbial profiling techniques, such as Lectin-seq (McPherson et al. 2023), where lectin-specific sorting combined with metagenomic sequencing provides a detailed landscape of microbial diversity at the host-microbe interface and its alterations in disease.

Finally, testing GAPs for mucin binding illustrated their utility in identifying host glycan signatures. This screening allowed for the prediction of interactions across various glycan types in complex biological environments where both microbial and host glycans coexist, such as the oral cavity and digestive tract. Additionally, GAP-mediated histochemistry provided a potential diagnostic and prognostic alternative to immunostaining in differentiating diseased vs healthy tissues.

Collectively, GAPs are valuable resources to probe the role of protein-glycan interactions and leverage them for diagnostic and therapeutic interventions. GAPs can be beneficial in building a comprehensive database of glycan epitopes involved in host–microbe interaction. The curated databases, combined with artificial intelligence methods like machine learning (Bojar et al. 2022), can predict unknown glycan-mediated interactions in host-microbe dynamics, advancing the development of glycoconjugate-based therapeutics. Additionally, GAPs can serve as a platform for developing proteins, peptide-based inhibitors, or small molecules that disrupt native lectin-glycan interactions, offering a promising strategy to block pathogen adhesion and infection.

Materials and methods

The recombinant production and characterization of GAPs and screening of GAPs on mammalian glycan arrays are discussed in detail in the supporting information text.

Screening of GAPs on custom bacterial glycan array

The slide printed with microbial glycans (from TSRI MicroArray Core Facility, The Scripps Research Institute) was equilibrated at room temperature after taking out of −20 °C for 20–30 min. The slide was placed on a slide holder tube and soaked in wash buffer (20 mM HEPES; pH 7.5, 150 mM NaCl, 10 mM $CaCl_2$) for 10 min. The slide was placed on a raised slide holder platform in a humidification chamber with wet paper towels and the printed area (no barcode side) was sealed by placing a rubber guard around it. 800 μL of Strep-II tagged lectin, diluted to 5 μg/mL (for Galectin-3,4,8), 10 μg/mL (for full-length MBL, and ΔN-SP-D) or 50 μg/mL (for full-length CL-K1 and PTX3) in protein binding buffer (20 mM HEPES; pH 7.5, 150 mM NaCl, 10 mM $CaCl_2$, 0.1% BSA, 0.1% Tween 20) was added on the slide and incubated for 1 h at room temperature with gentle rocking inside the humidification chamber. Biotinylated ZG16P (800 μL) at a concentration of 10 μg/mL (either free or pre-complexed with streptavidin-Cy3) in protein binding buffer (20 mM HEPES; pH 7.5, 150 mM NaCl, 10 mM $CaCl_2$, 0.1% BSA, 0.1% Tween 20) was added to the slides. Meanwhile, the Dy549-conjugated strep-monoclonal antibody (StrepMAB) was prepared by diluting the stock solution (from IBA-Lifesciences) of 0.5 mg/mL in 750 μL protein binding buffer at a final concentration of 4 ug/mL. For biotinylated GAPs, the Streptavidin conjugated Cy3 (RayBiotech 300 glycan array kit) was prepared according to the manufacturer's protocol. After the incubation, the slide was washed in protein binding buffer, wash buffer, and double-deionized water consecutively, four times in each wash medium. The slide was placed back in the chamber, sealed with the rubber guard and 750 μL of Dy549-conjugated strep-monoclonal antibody was added to the slide and incubated for 1 h at room temperature with gentle rocking. The slide was then washed in protein binding buffer and wash buffer consecutively, four times in each medium. Lastly, the slide was washed three times in double-deionized water in two separate wash containers. The slide was dried in a slide spinner and scanned at the Genepix4000 laser scanner using the 532 nm laser. The data obtained from the instrument were analyzed by using Microsoft Excel to average the fluorescence intensities of the best four of six replicates for each glycan. The average intensity was plotted against the sample ID number.

Mucin binding assay

Lyophilized MUC2, MUC5AC, and MUC5B were generously provided by Prof. Katharina Ribbeck, whose group purified them from porcine sources as previously described (Wu et al. 2023). These mucins were reconstituted to 10 μg/μL overnight at 4 °C in mucin assay buffer (20 mM HEPES, 150 mM sodium chloride, pH 7.4) as previously described (Diehl et al. 2024), then diluted to a concentration of 1 μg/μL in the same buffer. The porcine mucins Type II and type III (Sigma) were diluted at the same concentration and treated with 1 mM sodium periodate at 37 °C overnight, protected from light. One μg of the mucin samples was dotted on nitrocellulose or PVDF membranes in triplicate. The membranes were incubated with biotinylated or Strep-II-tagged GAPs at appropriate concentrations. For galectins, membranes were washed with mucin assay buffer supplemented with 50 mM lactose after incubating with GAPs. The blots were visualized with appropriate detection methods at different conditions (SI text) and analyzed using the Fiji image suite.

Fluorescence immunohistochemistry using GAP

Surgical tissue of mouse ileum and colon, fixed in Carnoy's fixative, were paraffin-embedded and sectioned (4–5 μM) and mounted on X-tra™ positive-charged slides (Leica Biosystems, Buffalo Grove, IL). Sections were deparaffinized, rehydrated in gradient ethanol, H_2O, and blocked 5% BSA (in 1 × PBS) for 60 min before overnight incubation at 4 °C with

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FITC-conjugated Galectin-3 GAP. Following overnight incubation with GAP, samples were counter-stained with DAPI (10 μg/mL) and mounted with ProLong Diamond antifade mountant (Thermo) with a 24-hour cure time. Images were captured using RPI spinning disk confocal microscope.

Flow cytometry analysis of stool samples

100 μl of stool homogenate was thawed on ice and washed three times with 1 mL of PBS-BSA-T [PBS (pH 7.4; Gibco), 0.1% (w/v) BSA (US Biological, A1311), and 0.05% (v/v) Tween 20 (Sigma-Aldrich)]. The samples were pelleted via centrifugation at 3,000 RCF for 10 min, and resuspended in PBS-BSA-T. The stool samples were then stained at OD_{600} of 0.2. Staining solutions were HEPES-Ca^{2+}-BSA-T with SYTO BC (1:1000; Thermo Fisher Scientific) and StrepMAB classic DY649-conjugate (1:250; IBA Lifesciences) or Alexa Fluor® 647 anti-His Tag Antibody (1:250; Biolegend) for the unstained controls. All StrepII-tagged lectin-stained Ca^{2+} samples were stained as follows: Hepes-Ca^{2+}-BSA-T, SYTO BC (1:1000), StrepMAB classic DY649-conjugate (1:250), and recombinant StrepII-tagged lectin (0.5 μM). All His-tagged lectin-stained samples were stained as follows: HEPES-Ca^{2+}-BSA-T, SYTO BC (1:1000), Alexa Fluor® 647 anti-His Tag Antibody (1:250; Biolegend), and recombinant His-tagged lectin (0.55 μM). For the lectin-stained EDTA control samples, the staining conditions were as follows: HEPES-EDTA-BSA-T [20 mM Hepes (pH 7.4), 150 mM NaCl, 1 mM EDTA, 0.1% BSA, and 0.05% Tween 20], SYTO BC (1:1000), StrepMAB classic DY649-conjugate (1:250) or Alexa Fluor® 647 anti-His Tag Antibody (1250; Biolegend) with tagged lectin (0.5 μM). Staining was performed at 4 °C for 1 h before being diluted 4× and analyzed on Attune Flow Cytometer (Thermo Fisher Scientific).

Human sample research

Human stool samples from healthy individuals were obtained under a protocol approved by the Institutional Review Board at the Massachusetts Institute of Technology (IRB no. 1510271631). Stool samples were collected from participants enrolled in the Prospective Registry in IBD Study at MGH (PRISM), with approval from the Partners Human Research Committee (protocol no. 2004-P-001067). All procedures were conducted following the respective institutional review board guidelines and the informed consent from the participants.

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Author contributions

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Supplementary material

Supplementary material is available at *Glycobiology Journal* online.

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Conflict of interest statement

None declared.

Data availability

The constructs used for developing GAPs are available to the community. Please contact Dr. Barbara Imperiali (imper@mit.edu) and Dr. Laura L. Kiessling (kiesslin@mit.edu) for the GAP constructs.

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